

ZAINAB IRFAN

AI/ML Engineer • RAG & LLM Systems Specialist • Full-Stack Python Developer

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PROFESSIONAL PROFILE

Motivated **AI/ML Engineer** with hands-on, project-driven experience designing and building **Retrieval-Augmented Generation (RAG) systems, conversational LLM agents, and full-stack AI platforms**. Has independently designed, built, and deployed a wide range of practice and personal projects — from data pipelines and model integration to production-style, user-facing applications — using **FastAPI, Next.js, LangChain**, and vector databases such as **Qdrant**. Participated in the **Alibaba Cloud AI Hackathon**, applying rapid prototyping and applied problem-solving skills under competitive, time-boxed conditions. Combines strong software engineering fundamentals with applied machine learning expertise, backed by **7 industry certifications** from AWS, IBM, and DeepLearning.AI.

CORE TECHNICAL EXPERTISE

- **AI / LLM Engineering** — RAG Pipeline Architecture · LangChain · Prompt Engineering · LLM Fine-Tuning (LoRA/QLoRA, PEFT) · LLM Evaluation & Benchmarking · Conversational AI Agents · Streaming Chat Interfaces · Workflow Automation (n8n)
- **Machine Learning** — Supervised & Unsupervised Learning · Feature Engineering · Model Evaluation · Bias Detection · Exploratory Data Analysis (EDA)
- **Languages & Frameworks** — Python · JavaScript · HTML/CSS · FastAPI · Streamlit · Next.js · React · PyTorch · Transformers
- **Data & Vector Infrastructure** — Qdrant · Supabase · NumPy · Pandas · Scikit-learn · Matplotlib · Seaborn
- **APIs, Cloud & Deployment** — REST API Design & Integration · AWS (AI/ML) · Render Deployment · Infrastructure Automation · Google Colab (NVIDIA T4 GPU)
- **Research & Problem-Solving** — Technical Research · Rapid Prototyping · Hackathon Problem-Solving · Independent Learning
- **Languages Spoken** — English (Proficient) · Urdu (Native)

EXPERIENCE & INVOLVEMENT

- **Alibaba Cloud AI Hackathon — Participant 2026**
- Collaborated under competitive, time-boxed conditions to design and prototype an AI-powered solution, sharpening rapid prototyping, teamwork, and applied problem-solving skills.
- **Independent Project Builder 2024 – Present**
- Self-directed development of 8+ full-stack AI/ML projects for hands-on practice — spanning RAG pipelines, LLM fine-tuning, conversational agents, and workflow automation — covering the complete lifecycle from API design and data pipelines to model integration and cloud deployment.

KEY PROJECTS

Green Flora — Smart Agriculture Platform 2026

Stack: Next.js, FastAPI, LLM APIs, REST APIs · Full-Stack Deployment

- Built a full-stack smart agriculture platform unifying an **AI farming assistant, weather insights**, an image-based **Crop Doctor**, **live market trends**, farm management, and a **profit calculator** into one farmer-friendly dashboard.
- Developed a **multilingual AI assistant** (English, Urdu, Roman Urdu) that answers farming questions via text or voice using live weather, market, and agricultural data.
- Engineered the frontend in **Next.js** and backend **REST APIs** in **FastAPI**, integrating real-time weather and market data with LLM-based crop analysis.

Skills Applied: Next.js, FastAPI, REST API Design, LLM Integration, Multilingual AI, Full-Stack Deployment

Prism AI — RAG-Based Document Intelligence Platform 2026

Stack: Next.js, FastAPI, LangChain, Qdrant, Supabase, OpenAI/Claude APIs · Deployed on Render

- Built a full-stack **Retrieval-Augmented Generation** platform that ingests multi-format documents (PDF, DOCX, PPT, CSV) and returns structured, context-aware answers.
- Designed the retrieval pipeline with **LangChain** and **Qdrant** for semantic search, using **Supabase** for persistent storage of documents and chat history.

- Integrated **LLM APIs** (OpenAI and Claude) with streamed chat responses for a real-time conversational Q&A experience.
- Built an **analytics dashboard** visualizing usage insights, queries per day, question volume, and engagement trends with interactive charts.
- Deployed a production build with a **Next.js** frontend and **FastAPI** backend hosted on **Render**.

Skills Applied: RAG Pipeline Design, LangChain, Vector Search (Qdrant), FastAPI, Next.js, LLM API Integration, Streaming Interfaces, Cloud Deployment, Data Visualization

Efficient LLM Fine-Tuning using LoRA & QLoRA (TinyLlama-1.1B-Chat) 2026

Stack: Python, PyTorch, Transformers, PEFT, TRL, BitsAndBytes, Hugging Face Datasets · Google Colab (NVIDIA T4 GPU)

- Fine-tuned **TinyLlama-1.1B-Chat** for Text-to-SQL generation using **LoRA** and **QLoRA**, on ~3,000 curated examples from the Hugging Face b-mc2/sql-create-context dataset.
- Trained on Google Colab with an **NVIDIA T4 GPU** across 3 epochs, reaching **84.2% validation accuracy**.
- Applied **4-bit quantization** and low-rank adaptation to substantially cut GPU memory usage and training cost while preserving model performance.

Skills Applied: LLM Fine-Tuning, LoRA, QLoRA, PEFT, 4-bit Quantization, GPU Optimization, Hugging Face, PyTorch, Text-to-SQL

AI Customer Support Agent for Shopify Stores 2025 – 2026

Stack: Python, LLM APIs, Shopify Integration

- Developed an **AI-powered customer support agent** that integrates with a Shopify storefront to handle customer queries automatically.
- Used **LLM APIs** to generate context-aware responses, reducing manual reply workload for store operations.

Skills Applied: LLM API Integration, Conversational AI Agents, Third-Party API Integration, Python

Automated Email Reply System 2025

Stack: Python, LLM APIs, Email Automation, n8n

- Built an **automation system** that reads incoming emails, generates context-appropriate draft replies using an LLM, and sends responses automatically.

Skills Applied: Workflow Automation (n8n), LLM Prompt Engineering, Python, Email/API Integration

Comprehensive Clinical Data Analysis & Visualization 2025

Stack: Python, Pandas, Matplotlib/Seaborn, Scikit-learn

- Performed end-to-end **exploratory data analysis (EDA)** on raw clinical datasets, including cleaning, transformation, and feature engineering.
- Built visualizations of clinical insights such as patient distribution by disease and other key trends across the dataset.
- Applied **machine learning techniques** to generate predictive insights from patient records.

Skills Applied: Exploratory Data Analysis, Feature Engineering, Pandas, Data Visualization (Matplotlib/Seaborn), Scikit-learn, Predictive Modeling

CERTIFICATIONS

- **AI Fundamentals & Cloud** — AWS
- **AI Workflow: Feature Engineering & Bias Detection** — IBM
- **Building Systems with the ChatGPT API** — DeepLearning.AI
- **Machine Learning for All** — University of London
- **Machine Learning** — Simplilearn
- **App Deployment & Security** — Packt
- **Learn to Code with AI** — Scrimba